C O N F I D E N T I A L

Statement of Work

Xponent Technologies proposes the following project to Huzaifa Ali to develop a timeline and cost proposal.

Project Description

Standalone development of a microphone solution that will act as part of a sleep mask. The goal is to pick up environmental sounds that may be the source of sleep disturbances but also possible sounds by the user, such as snoring.
The project does not aim at the development of an entire PCB, but only at the microphone subsystem to be fed to our main PCB designer. The prototype utilizes a simple processor package like ESP32 so we can read raw data for testing the hardware.

(A) Deliverables
(1) A block diagram
(2) Picking microphone(s) with reliability of supply and large stock
(3) Quick prototype to confirm the microphone setup and the threshold triggering work (provide data remotely, then send prototype if satisfied)

(B) For context, we are providing specific user requirements and design inputs that your project will at least partially cater to but are not meant as project deliverables or KPIs.

User Requirements

- The prototype firmware needs to be easily and rapidly updated
- Data recorded from the prototype needs to be transmitted conveniently
- Emily needs to be able to understand the reasons for poor sleep
- Emily needs to be able to listen to sounds while sleeping comfortably and personally (without triggering the mic)
- Emily needs to be able to understand reasons for unwanted waking up at night
- Emily needs to be able to optimize their sleep environment
- The microphone system needs to be energy efficient

Design Inputs

- No electronics can be seen from the outside (mic behind channel in fabric or layer of fabric)
- Exposed electronics (not in the electronics pouch) like the charge port, buttons and microphone are non-moisture sensitive
- The electronics do not have sharp or hard edges that would pierce the fabric
- Audio recording is not triggered by sounds played by the mask speakers
- The electrical system of the device should have enough power to last for 27-63 hours of active wear time (be very power conservative)
- The system can detect and record noises in the environment and can trigger events if they meet specific criteria (i.e. dB threshold)
- The microphone system needs to be reliable at recording when the threshold trigger is hit, but no unnecessary data is recorded
- All materials and components used to have RoHS documentation available
- All materials and components have long-term stock available and the supply chain is transparent and reliable (we have a supplier evaluation process)

(C) Next Steps
Please estimate the hours/cost for each deliverable and a reasonable estimated target date of completion, assuming a project start on Jan 31.